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Suppression of inter-modal four-wave mixing in high-power fiber lasers

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Abstract: For the inter-modal four-wave mixing (IMFWM) in high-power fiber lasers, the phase-matching frequency shift and coherence length are calculated to determine the fiber mode combinations corresponding to IMFWM peaks; then, the parameters of the laser are optimized accordingly to suppress the IMFWM. On the basis of laser parameter optimization, the fiber coiling method is applied to further suppress the IMFWM. To validate this method, a master oscillator power amplifier was configured with 20/400 fiber to produce IMFWM peaks at 1108 and 1071.6 nm in the output spectrum, and then those peaks were removed by reducing the bending radius of the fiber.

OCIS codes: (140.3510) Lasers, fiber; (140.3460) Lasers; (190.4370) Nonlinear optics, fibers.

References and links

1. D. J. Richardson, J. Nilsson, and W. A. Clarkson, "High power fiber lasers: current status and future perspective," J. Opt. Soc. Am. B **27**(11), B63–B92 (2010).
2. J. Nilsson and D. N. Payne, "Physics. High-power fiber lasers," Science **332**(6032), 921–922 (2011).
3. R. H. Stolen, J. E. Bjorkholm, and A. Ashkin, "Phase-matched three-wave mixing in silica fiber optical waveguides," Appl. Phys. Lett. **24**(7), 308–310 (1974).
4. R. H. Stolen, "Phase-matched-stimulated four-photon mixing in silica-fiber waveguides," IEEE. J. Quantum Electron. **11**(3), 100–103 (1975).
5. C. Lin and M. A. Bösch, "Large-Stokes-shift stimulated four-photon mixing in optical fibers," Appl. Phys. Lett. **38**(7), 79–481 (1981).
6. P. L. Baldeck and R. R. Alfano, "Intensity effects on the stimulated four photon spectra generated by picosecond pulses in optical fibers," J. Lightwave Technol. **5**(12), 1712–1715 (1987).
7. H. Pourbeyram, E. Nazemosadat, and A. Mafi, "Detailed investigation of intermodal four-wave mixing in SMF-28: blue-red generation from green," Opt. Express **23**(11), 14487–14500 (2015).
8. S. M. M. Friis, I. Begleris, Y. Jung, K. Rottwitt, P. Petropoulos, D. J. Richardson, P. Horak, and F. Parmigiani, "Inter-modal four-wave mixing study in a two-mode fiber," Opt. Express **24**(26), 30338–30349 (2016).
9. R. Dupiol, A. Bendahmane, K. Krupa, A. Tonello, M. Fabert, B. Kibler, T. Sylvestre, A. Barthelemy, V. Couderc, S. Wabnitz, and G. Millot, "Far-detuned cascaded intermodal four-wave mixing in a multimode fiber," Opt. Lett. **42**(7), 1293–1296 (2017).
10. J. Cheng, M. E. Pedersen, K. Charan, K. Wang, C. Xu, L. Grüner-Nielsen, and D. Jakobsen, "Intermodal four-wave mixing in a higher-order-mode fiber," Appl. Phys. Lett. **101**(16), 161106 (2012).
11. S. R. Petersen, T. T. Alkeskjold, C. B. Olausson, and J. Lægsgaard, "Intermodal and cross-polarization four-wave mixing in large-core hybrid photonic crystal fibers," Opt. Express **23**(5), 5954–5971 (2015).
12. A. Labruyère, A. Martin, P. Leproux, V. Couderc, A. Tonello, and N. Traynor, "Controlling intermodal four-wave mixing from the design of microstructured optical fibers," Opt. Express **16**(26), 21997–22002 (2008).
13. R. J. Essiambre, M. A. Mestre, R. Ryf, A. H. Gnauck, R. W. Tkach, A. R. Chraplyvy, Y. Sun, X. L. Jiang, and R. Lingle, "Experimental investigation of inter-modal four-wave mixing in few-mode fibers," IEEE Photonics Technol. Lett. **25**(6), 539–542 (2013).
14. Y. Xiao, R. J. Essiambre, M. Desgroseilliers, A. M. Tulino, R. Ryf, S. Mumtaz, and G. P. Agrawal, "Theory of intermodal four-wave mixing with random linear mode coupling in few-mode fibers," Opt. Express **22**(26), 32039–32059 (2014).
15. Z. Q. Pan, Y. Weng, and X. He, "Investigation of the nonlinearity in few mode fibers," in *Proceedings of IEEE Conference* on International Conference on Optical Communications and Networks (IEEE, 2014), pp. 1–4.
16. M. A. Lapointe and M. Piché, "Linewidth of high-power fiber lasers," Proc. SPIE **7386**, 73860S (2009).

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17. Y. J. Feng, X. J. Wang, W. W. Ke, Y. H. Sun, K. Zhang, Y. Ma, T. L. Li, Y. S. Wang, and J. Wu, "Numerical analysis to four-wave mixing induced spectral broadening in high power fiber lasers," Proc. SPIE **9255**, 92550Q (2015).
18. M. Hu, W. W. Ke, Y. F. Yang, M. Lei, K. Liu, X. L. Chen, C. Zhao, Y. F. Qi, B. He, X. J. Wang, and J. Zhou, "Low threshold Raman effect in high power narrowband fiber amplifier," Chin. Opt. Lett. **14**(1), 011901 (2016).
19. Z. B. Li, Z. H. Huang, X. Y. Xiang, X. B. Liang, H. H. Lin, S. H. Xu, Z. M. Yang, J. J. Wang, and F. Jing, "Experimental demonstration of transverse mode instability enhancement by a counter-pumped scheme in a 2 kW all-fiberized laser," Photon. Res. **5**(2), 77–81 (2017).
20. Q. Fang, J. H. Li, W. Shi, Y. G. Qin, Y. Xu, X. J. Meng, R. A. Norwood, and N. Peyghambarian, "5kW Near-diffraction-limited and 8kW High-brightness Monolithic Continuous Wave Fiber Lasers Directly Pumped by Laser Diodes," IEEE Photonics J. **9**(5), 1506107 (2017).
21. J. P. Koplow, D. A. V. Kliner, and L. Goldberg, "Single-mode operation of a coiled multimode fiber amplifier," Opt. Lett. **25**(7), 442–444 (2000).
22. G. P. Agrawal, *Nonlinear Fiber Optics* (Academic, 2007).
23. D. Gloge, "Weakly guiding fibers," Appl. Opt. **10**(10), 2252–2258 (1971).
24. C. L. Chen, *Foundations for Guided-Wave Optics* (John Wiley & Sons, 2006).
25. J. Wang, D. Yan, S. Xiong, B. Huang, and C. Li, "High power all-fiber amplifier with different seed power injection," Opt. Express **24**(13), 14463–14469 (2016).
26. Y. Xu, Q. Fang, Y. Qin, X. Meng, and W. Shi, "2 kW narrow spectral width monolithic continuous wave in a near-diffraction-limited fiber laser," Appl. Opt. **54**(32), 9419–9421 (2015).
27. I. H. Malitson, "Interspecimen Comparison of the Refractive Index of Fused Silica," J. Opt. Soc. Am. **55**(10), 1205–1208 (1965).
28. H. Yu, H. Zhang, H. Lv, X. Wang, J. Leng, H. Xiao, S. Guo, P. Zhou, X. Xu, and J. Chen, "3.15 kW direct diode-pumped near diffraction-limited all-fiber-integrated fiber laser," Appl. Opt. **54**(14), 4556–4560 (2015).
29. D. Marcuse, "Curvature loss formula for optical fibers," J. Opt. Soc. Am. **66**(3), 216–220 (1976).
30. R. T. Schermer and J. H. Cole, "Improved Bend Loss Formula Verified for Optical Fiber by Simulation and Experiment," IEEE. J. Quantum Electron. **43**(10), 899–909 (2007).
31. M. Gong, Y. Yuan, C. Li, P. Yan, H. Zhang, and S. Liao, "Numerical modeling of transverse mode competition in strongly pumped multimode fiber lasers and amplifiers," Opt. Express **15**(6), 3236–3246 (2007).

1. Introduction

Fiber lasers are presently applied in a wide range of fields, including optical communications, military defense, and industrial processing, because they provide the benefits of high efficiency, high spatial beam quality, compactness, and reliability [1,2]. In high-power fiber lasers, the application of large-mode-area (LMA) double-clad fiber (DCF) is the most direct and effective means to reduce the power density of the fiber core and improve the output laser power. However, the consequence of increasing the size of the fiber core is that more spatial modes are produced, which results in inter-modal four-wave mixing (IMFWM). Four-wave mixing (FWM) is a process wherein two photons are annihilated to produce Stokes and anti-Stokes photons, and can occur when the phase-matching condition is satisfied. In multimode fibers, when fiber modes in which four waves participating in the FWM process propagate are different, this process is referred to as IMFWM. The IMFWM peaks, i.e., the Stokes and anti-Stokes peaks generated by the IMFWM process, corresponding to the different combinations of fiber modes for which phase matching is achieved, can be observed in the output spectrum. The greater the number of modes supported by the fiber, the greater the number of Stokes and anti-Stokes peaks produced by the IMFWM process. The presence of these peaks results in deterioration of the output spectrum of a high-power fiber laser, which limits its application in the field of spectral beam combination. And these IMFWM peaks also causes significant energy dispersion of the output laser, which limits its maximum obtainable output power. For these reasons, it is important to research IMFWM to find ways to overcome these limitations in the production and operation of high-power fiber lasers and amplifiers.

Research on IMFWM began in the 1970s. Stolen et al. exhibited IMFWM for the first time in silica-based fibers at visible wavelengths by achieving phase matching between different spatial modes [3,4]. Then, Lin et al. reported the first observation of large-frequency-shift (2000 to over 4000 cm^{-1}) IMFWM in low-mode-number silica fibers [5]. In 1987, Baldeck et al. used a picosecond pump pulse propagated through a multimode fiber supporting four modes to generate a supercontinuum extending from 530 to 580 nm [6]. In recent years, IMFWM has been studied in few-mode fibers (FMFs), higher-order-mode fibers, photonic crystal fibers, and microstructured optical fibers [7–12]. In the field of fiber communications in particular, Essiambre et al. experimentally demonstrate nondegenerate IMFWM between waves belonging to different spatial modes of a 5-km-long FMF [13]. Then, Xiao et al. researched the impact of random linear mode coupling on IMFWM [14]. Pan et al. analyzed the effects of differential mode group delay, random mode coupling, and wavelength separation upon IMFWM efficiency in FMFs [15]. These results provide suitable theoretical guidelines on which to base research on IMFWM in high-power fiber lasers.

In the field of high-power fiber lasers, researchers have primarily focused on analyzing the effects of FWM on the spectral broadening of fiber lasers. Lapointe et al. pointed out that FWM between multiple longitudinal modes results in the spectral broadening of a fiber laser oscillator with fiber Bragg gratings (FBGs) [16]. Feng et al. proposed a numerical calculation model of the spectral broadening induced by FWM and pointed out that shortening the fiber or using an LMA fiber would decrease the extent of the broadening to some degree [17]. In the literature on IMFWM in high-power fiber lasers [18] and [19], the Stokes and anti-Stokes peaks caused by IMFWM were observed in the output spectra of fiber lasers; however, no detailed explanation as to how those peaks were generated was provided. In 2017, Fang et al. observed peaks around 1060 nm and 1100 nm in the output spectrum, and indicated they were the result of FWM in the fiber amplifier due to phase matching between the fundamental and high-order modes [20]. However, no detailed theoretical and experimental research on IMFWM has been performed to date.

In order to suppress the IMFWM in high-power fiber lasers, we propose a calculation model of the phase-matching frequency shift and coherence length to determine the fiber mode combination corresponding to each pair of IMFWM peaks observed in the output spectrum. Then, based on the calculation model, the influence of various parameters, such as, the fiber core radius, numerical aperture, and light wavelength, on the phase-matching frequency shift and the number of pairs of IMFWM peaks is analyzed to provide theoretical guidance for optimizing the laser parameters to suppress the IMFWM. However, considering the limitations of laser parameter optimization, the fiber coiling method [21], which is well-known for high-order modes (HOMs) suppression, is applied to further suppress the IMFWM. And a simulation model of the fiber coiling method is established to demonstrate the efficacy of HOMs suppression and to provide theoretical guidance for the selection of the fiber bending radius. To validate this approach, we constructed a fiber laser system based on a master oscillator power amplifier (MOPA) configuration and mapped the resulting fiber mode combinations to the IMFWM peaks observed in the output spectrum according to the theoretical model. As per the results of the simulation model of the fiber coiling method, the bending radius of the gain fiber in the fiber amplifier was modified to verify the suppression of the IMFWM.

2. Principle

2.1 IMFWM in high-power fiber lasers

In high-power fiber lasers, the application of LMA DCF is the most direct and effective means to improve the output laser power. However, increasing the size of the fiber core results in more spatial modes, thus resulting in IMFWM. When two waves at frequencies ω_1 and ω_2 propagate in a multimode fiber and the phase-matching condition is satisfied, two waves at new frequencies ω_3 and ω_4 are generated. Assuming $\omega_3 < \omega_4$, the low-frequency wave at ω_3 and the high-frequency wave at ω_4 are referred to as the Stokes and anti-Stokes waves, respectively. If the four waves at frequencies ω_1, ω_2, ω_3, and ω_4 are in the same fiber modes, this process is called the FWM process. In contrast, if the four waves at frequencies ω_1, ω_2, ω_3, and ω_4 are in different fiber modes, this process is referred to as the IMFWM process. A comparison between FWM and IMFWM in a high-power fiber laser is shown in Fig. 1.

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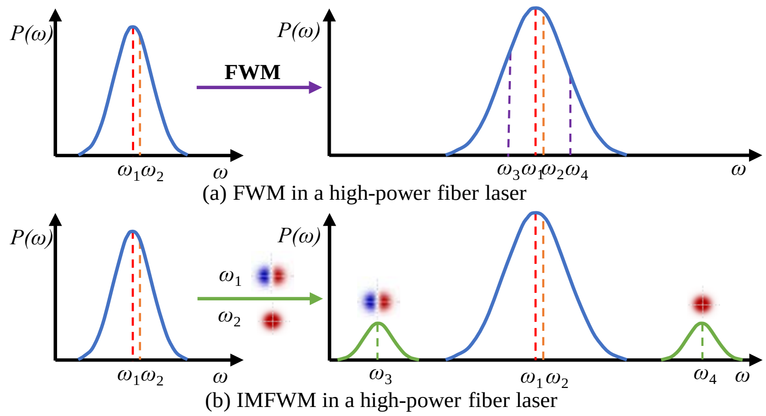

Fig. 1. Schematic diagram of a comparison between the FWM and IMFWM in a high-power fiber laser.

As shown in Fig. 1(a), taking the fiber laser based on the MOPA scheme as an example, when two waves at frequencies ω_1 and ω_2 are launched into the fiber amplifier as seed light, the FWM process causes the energy of the two waves at ω_1 and ω_2 to transfer to the two waves at ω_3 and ω_4. Because significant FWM occurs only if the phase mismatch is small, i.e., the frequency shift $\Omega_s = \omega_1 - \omega_3 = \omega_4 - \omega_2$ that satisfies the phase-matching condition is small, the FWM process causes the conversion of energy between multiple wavelengths, resulting in the spectral broadening of the fiber laser. As shown in Fig. 1(b), when the wave at frequency ω_1 is in the LP_{11} mode and the wave at frequency ω_2 is in the LP_{01} mode, the phase-matching frequency shift of the IMFWM process is much larger than that of the FWM process. This is because the magnitude of the phase mismatch occurring as a result of the waveguide dispersion depends on the choice of fiber modes in which the four waves participating in the IMFWM process propagate. The Stokes peak at frequency ω_3 and anti-Stokes peak at frequency ω_4, which are produced by IMFWM, can be observed in the output spectrum of the high-power fiber laser. As the phase-matching frequency shifts corresponding to different fiber mode combinations are different, the greater the number of fiber modes supported in the fiber, the more Stokes and anti-Stokes peaks can be generated by the IMFWM process. This results in deterioration of the output spectrum of the fiber laser and significant energy dispersion of the output laser, which limits its maximum obtainable output power and its application in the field of spectral beam combination. This is why it is important to study how to best suppress the generation of IMFWM in high-power fiber lasers.

2.2 Calculation model of the phase-matching frequency shift and coherence length

In order to suppress the generation of IMFWM products in high-power fiber lasers, we must first determine the fiber mode combination corresponding to each pair of IMFWM peaks observed in the output spectrum. That is, we need to calculate the phase-matching frequency shift corresponding to different fiber mode combinations. The phase-matching frequency shift is the frequency shift Ω_s that satisfies the phase-matching condition $\kappa = 0$, where κ is the phase mismatch and can be written as

$$\kappa = \Delta k_M + \Delta k_W + \Delta k_{NL} = 0. \tag{1}$$

Here, Δk_M, Δk_W, and Δk_{NL} represent the mismatches cause $\kappa = \Delta k_M + \Delta k_W + \Delta k_{NL} = 0.$ d by material dispersion, waveguide dispersion, and nonlinear effects, respectively [22]. The magnitude of the waveguide contribution Δk_W depends on the choice of fiber modes in which the four waves participating in the IMFWM process propagate. When the waveguide contribution Δk_W is negative and exactly compensates for the positive contribution Δk_W +

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Δk_{NL}, the phase matching condition is satisfied and IMFWM products can therefore be generated.

Assuming four waves participate in the IMFWM process at frequencies ω_1, ω_2, ω_3, and ω_4, if the effective indices at frequency ω_j are written as $\tilde{n}_j = n_j + \Delta n_j$ $(j = 1\ to\ 4)$, the material contribution Δk_M and waveguide contribution Δk_W are

$$\begin{aligned} \Delta k_M &= (n_3\omega_3 + n_4\omega_4 - n_1\omega_1 - n_2\omega_2)/c \\ \Delta k_W &= (\Delta n_3\omega_3 + \Delta n_4\omega_4 - \Delta n_1\omega_1 - \Delta n_2\omega_2)/c, \end{aligned} \tag{2}$$

where Δn_j is the change in the material index due to waveguiding. Considering the propagation constant $\beta(\omega_j) = n_j\omega_j / c$, the material contribution Δk_M can be written as

$$\Delta k_M = \beta(\omega_3) + \beta(\omega_4) - \beta(\omega_1) - \beta(\omega_2). \tag{3}$$

Considering the frequency shift $\Omega_s = \omega_1 - \omega_3 = \omega_4 - \omega_2$, and respectively expanding the propagation constants $\beta(\omega_3)$ and $\beta(\omega_4)$ in a Taylor series around the frequencies ω_1 and ω_2 while retaining up to the fourth-order terms in Ω_s, the material contribution Δk_M can be expressed as

$$\begin{aligned} \Delta k_M \approx{} & \Omega_s(\beta_1(\omega_2)\text{-}\beta_1(\omega_1)) + \Omega_s^2(\beta_2(\omega_1) + \beta_2(\omega_2))/2 + \Omega_s^4(\beta_3(\omega_2)\text{-}\beta_3(\omega_1))/6 \\ & + \Omega_s^4(\beta_4(\omega_1) + \beta_4(\omega_2))/24 \end{aligned}. \tag{4}$$

If the light wavelengths ($\lambda_1 = 2\pi c/\omega_1$, $\lambda_2 = 2\pi c/\omega_2$) are not too close to the zero-dispersion wavelength of the fiber, the high-order terms in Ω_s can be neglected, and Eq. (4) can be approximated as

$$\Delta k_M \approx \Omega_s(\beta_1(\omega_2)\text{-}\beta_1(\omega_1)) + \Omega_s^2(\beta_2(\omega_1) + \beta_2(\omega_2))/2. \tag{5}$$

Here, $\beta_1(\omega_1)$ and $\beta_2(\omega_1)$ are the dispersion parameters at frequency ω_1, and $\beta_1(\omega_2)$, $\beta_2(\omega_2)$ are the dispersion parameters at frequency ω_2. As $\Delta\overline{v} = \overline{v}_1 - \overline{v}_3 = \overline{v}_4 - \overline{v}_2$ is the frequency shift in units of cm^{-1} and $\overline{v} = \omega / 2\pi c$, the frequency shift Ω_s is $2\pi c\Delta\overline{v}$. Then, the material contribution Δk_M can be written as

$$\Delta k_M = 2\pi c(\beta_1(\overline{v}_2)\text{-}\beta_1(\overline{v}_1))\Delta\overline{v} + 2\pi^2 c^2(\beta_2(\overline{v}_1) + \beta_2(\overline{v}_2))\Delta\overline{v}^2. \tag{6}$$

For the waveguide contribution Δk_W, because $\omega = 2\pi c\overline{v}$, Eq. (2) can be rewritten as

$$\Delta k_W = 2\pi(\Delta n_3\overline{v}_3 + \Delta n_4\overline{v}_4 - \Delta n_1\overline{v}_1 - \Delta n_2\overline{v}_2). \tag{7}$$

For weakly guiding fibers with a small index difference, the normalized propagation constant can be approximated as [23]

$$b = \frac{(\beta/k)^2 - n^2}{n_c^2 - n^2} \approx \frac{\beta/k - n}{n_c - n} = \frac{\tilde{n} - n}{n_c - n}, \tag{8}$$

where k is the wave number and n_c, n are the refractive indices of the core and cladding, respectively. Substituting Eq. (8) into Eq. (7), we obtain the waveguide contribution

$$\Delta k_W = 2\pi(n_c - n)(b_3\overline{v}_3 + b_4\overline{v}_4 - b_1\overline{v}_1 - b_2\overline{v}_2). \tag{9}$$

The normalized propagation constant b can also be written as

$$b = \frac{\beta^2 - n^2k^2}{n_c^2k^2 - n^2k^2} = \frac{W^2}{V^2} = 1 - \frac{U^2}{V^2}. \tag{10}$$

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Here, $V = ka\sqrt{n_c^2 - n^2}$ is the normalized frequency, $W = a\sqrt{\beta^2 - k^2n^2}$ is the normalized radial attenuation constant, and $U = a\sqrt{k^2n_c^2 - \beta^2}$ is the normalized radial phase constant, where a is the core radius. The normalized propagation constant b for each fiber mode can be obtained as a function of the normalized frequency V based on the characteristic equation of each mode.

The fiber mode combinations that result in IMFWM can be separated into two cases. In Case 1, the waves at ω_1 and ω_2 are in different fiber modes, and the waves at ω_3 and ω_4, respectively, correspond to the fiber modes of the waves at ω_1 and ω_2. For example, ω_1: LP_{11}, ω_2: LP_{01}, ω_3: LP_{11}, and ω_4: LP_{01}, which can be written as (11,01→11,01). In Case 2, the waves at ω_1 and ω_2 are in the same fiber modes, but the waves at ω_3 and ω_4 are in different fiber modes. For example, ω_1: LP_{01}, ω_2: LP_{01}, ω_3: LP_{01}, and ω_4: LP_{11}. First, we derive an equation describing the waveguide contribution Δk_W for Case 1. Assuming that the fiber mode combination is (11,01→11,01), the waveguide contribution Δk_W can be expressed as

$$\Delta k_W = 2\pi(n_c - n)(b_{11}\bar{\nu}_3 + b_{01}\bar{\nu}_4 - b_{11}\bar{\nu}_1 - b_{01}\bar{\nu}_2). \tag{11}$$

Expanding $b_{11}\bar{\nu}_3$ about $\bar{\nu}_1$ and neglecting the high order terms in $\Delta\bar{\nu}$, we obtain

$$b_{11}\bar{\nu}_3 \approx b_{11}\bar{\nu}_1 + \left.\frac{d(b_{11}\bar{\nu})}{d\bar{\nu}}\right|_{\nu=\bar{\nu}_1}(\bar{\nu}_3 - \bar{\nu}_1) + \frac{1}{2}\left.\frac{d^2(b_{11}\bar{\nu})}{d\bar{\nu}^2}\right|_{\nu=\bar{\nu}_1}(\bar{\nu}_3 - \bar{\nu}_1)^2. \tag{12}$$

Because the wave number k is $2\pi\bar{\nu}$ and the normalized frequency V is $ka\sqrt{2\Delta}n_c$, where Δ is the relative refractive index difference, we obtain

$$\frac{d(b\bar{\nu})}{d\bar{\nu}} = \frac{1}{a\sqrt{2\Delta}n_c}\frac{d(Vb)}{dV}\frac{dV}{dk}. \tag{13}$$

As $\Delta = (n_c^2 - n^2)/2n_c^2 \approx (n_c - n)/n_c$, $d\Delta/dk = (n/n_c)[(1/n_c)(dn_c/dk) - (1/n)(dn/dk)]$. Since the core and cladding are made of the same basic materials, the dispersions of the core and cladding materials are similar [24]. Thus, $d\Delta/dk$ is small enough that it can be neglected, and due to $dn/d\lambda = (-2\pi/\lambda^2)(dk/d\lambda)$, we obtain $dV/dk \approx (V/k)[1-(\lambda/n_c)(dn_c/d\lambda)]$. In the near-infrared regions, $|(\lambda/n)(dn/d\lambda)|$ of most silica glass materials is on the order of 0.01 or smaller [24]. By ignoring the second term, we obtain $dV/dk \approx V/k$. Substituting this into Eq. (13), we obtain $d(b\bar{\nu})/d\bar{\nu} \approx d(Vb)/dV$. Substituting this into Eq. (12), and as $\bar{\nu} = V/2\pi a\sqrt{2\Delta}n_c$, we obtain

$$b_{11}\bar{\nu}_3 = b_{11}\bar{\nu}_1 - \frac{d(b_{11}V)}{dV}\Delta\bar{\nu} + \pi a\sqrt{2\Delta}n_c\frac{d^2(b_{11}V)}{dV^2}\Delta\bar{\nu}^2. \tag{14}$$

Similarly, expanding $b_{01}\bar{\nu}_4$ about $\bar{\nu}_2$ results in

$$b_{01}\bar{\nu}_4 = b_{01}\bar{\nu}_2 + \frac{d(b_{01}V)}{dV}\Delta\bar{\nu} + \pi a\sqrt{2\Delta}n_c\frac{d^2(b_{01}V)}{dV^2}\Delta\bar{\nu}^2. \tag{15}$$

Substituting Eqs. (14) and (15) into Eq. (11), the waveguide contribution can be expressed as

$$\Delta k_W = 2\pi(n_c - n)\left(\left(\frac{d(b_{01}V)}{dV} - \frac{d(b_{11}V)}{dV}\right)\Delta\bar{\nu} + \pi a\sqrt{2\Delta}n_c\left(\frac{d^2(b_{01}V)}{dV^2} + \frac{d^2(b_{11}V)}{dV^2}\right)\Delta\bar{\nu}^2\right). \tag{16}$$

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Similarly, we now derive an equation describing the waveguide contribution Δk_W for Case 2. Assuming that the fiber mode combination is (01,01→01,11), the waveguide contribution Δk_W can be expressed as

$$\begin{aligned}\Delta k_W &= 2\pi(n_c-n)(b_{01}\overline{\nu}_3+b_{11}\overline{\nu}_4-b_{01}\overline{\nu}_1-b_{01}\overline{\nu}_2)\\ &= 2\pi(n_c-n)\left((b_{11}-b_{01})\overline{\nu}_2+\left(\frac{d(b_{11}V)}{dV}-\frac{d(b_{01}V)}{dV}\right)\Delta\overline{\nu}+\pi a\sqrt{2\Delta}n_c\left(\frac{d^2(b_{01}V)}{dV^2}+\frac{d^2(b_{11}V)}{dV^2}\right)\Delta\overline{\nu}^2\right).\end{aligned} \tag{17}$$

Neglecting the contribution of nonlinear effects Δk_{NL}, the phase-matching condition is $\Delta k_M + \Delta k_W = 0$. The material contribution Δk_M and the waveguide contribution Δk_W related to the frequency shift $\Delta\overline{\nu}$ can be calculated according to Eq. (6) and Eqs. (16) and (17). We can then plot the curves of Δk_M and $-\Delta k_W$ versus the frequency shift, the intersection of which is the phase-matching frequency shift of the IMFWM process. Therefore, according to the above theoretical analysis, the phase-matching frequency shifts corresponding to different fiber mode combinations can be obtained by selecting the corresponding Δk_W formula.

In high-power fiber lasers, it is common to use at least tens of meters of fiber, and due to the fluctuation in the core radius, this will result in phase mismatch. Significant IMFWM can occur if the phase matching is not close enough to perfect to yield $\kappa = 0$. The occurrence of IMFWM products depends on the length of the fiber relative to the coherence length L_{coh}. Significant IMFWM products can occur when the fiber length satisfies the condition $L < L_{coh}$. According to the phase-matching condition, the coherence length is defined as

$$L_{coh} = 2\pi / |\delta\kappa| = 2\pi / |\delta(\Delta k_M(\Delta\overline{\nu})+\Delta k_W(\Delta\overline{\nu}))|, \tag{18}$$

where $|\delta\kappa|$ is the value of the phase mismatch due to the fluctuation in the core radius.

For Case 1, according to Eqs. (6) and (16), we have

$$\begin{aligned}\delta\kappa &= 2\pi(n_c-n)\Delta\overline{\nu}\delta a\left(\left(\frac{d^2(b_{01}V)}{dV^2}-\frac{d^2(b_{11}V)}{dV^2}\right)\frac{V}{a}+\pi\sqrt{2\Delta}n_c\left(\frac{d^2(b_{11}V)}{dV^2}+\frac{d^2(b_{01}V)}{dV^2}\right)\Delta\overline{\nu}\right)\\ &\quad +2\pi(n_c-n)\pi\sqrt{2\Delta}n_c V\left(\frac{d^3(b_{11}V)}{dV^3}+\frac{d^3(b_{01}V)}{dV^3}\right)\Delta\overline{\nu}^2\delta a \qquad ,\end{aligned} \tag{19}$$

where δa is the fluctuation in the core radius a.

Similarly, for Case 2, according to Eqs. (6) and (17), we have

$$\begin{aligned}\delta\kappa &= 2\pi(n_c-n)\frac{V}{a}\delta a\left(\frac{d(b_{11}-b_{01})}{dV}\overline{\nu}_2+\left(\frac{d^2(b_{11}V)}{dV^2}-\frac{d^2(b_{01}V)}{dV^2}\right)\Delta\overline{\nu}\right)\\ &\quad +2\pi(n_c-n)\pi\sqrt{2\Delta}n_c\Delta\overline{\nu}^2\delta a\left(\left(\frac{d^2(b_{01}V)}{dV^2}+\frac{d^2(b_{11}V)}{dV^2}\right)+V\left(\frac{d^3(b_{01}V)}{dV^3}+\frac{d^3(b_{11}V)}{dV^3}\right)\right).\end{aligned} \tag{20}$$

Therefore, the coherence lengths corresponding to different fiber mode combinations can be obtained according to Eqs. (18), (19), and (20), and the IMFWM products of different fiber mode combinations can occur when the fiber length of the high-power fiber laser is less than the corresponding coherence length.

3. Calculation and analysis

3.1 Calculation results of the phase-matching frequency shift and coherence length

As an example, considering the 25/400 μm (diameter of the core/inner cladding) fiber used in Refs [25-26], we calculated the phase-matching frequency shifts and coherence lengths corresponding to different fiber mode combinations as per the calculation model. For the 25/400 fiber, the core radius a is 12.5 μm and the numerical aperture NA_{co} of the core is 0.06. The refractive index of the inner cladding n is equal to that of pure fused silica, and it is related to the light wavelength [27]. When the light wavelength is 1080 nm, the refractive index of the inner cladding n is 1.4494, and the refractive index n_c of the core is calculated as $n_c = \sqrt{n^2 + NA_{co}^{\ 2}}$. The normalized frequency V of the 25/400 fiber is 4.36, which means that this fiber can support four modes (LP_{01}, LP_{11}, LP_{21}, and LP_{02}).

To simplify the calculation process, assuming that $\overline{\nu}_1 = \overline{\nu}_2$, the material contribution Δk_M can be simplified as

$$\Delta k_M = 4\pi^2 c^2 \beta_2(\overline{\nu}_1)\Delta\overline{\nu}^2, \tag{21}$$

where the dispersion parameter $\beta_2(\overline{\nu}_1)$ of fused silica for a wavelength of 1080 nm is 15.293 fs^2/mm. The curve of Δk_M versus the frequency shift according to Eq. (21) is plotted in Fig. 2, and the abscissa is transformed into $\Delta\nu = c\Delta\overline{\nu}$ for easier understanding, as indicated by the red dashed curves in Fig. 2.

Considering the three lower order modes LP_{01}, LP_{11}, and LP_{21}, for Case 1, there are three possible fiber mode combinations: (11,01→11,01), (21,01→21,01), and (21,11→21,11). According to Eq. (16), the blue solid line in Fig. 2(a) is the waveguide contribution Δk_W for (11,01→11,01) plotted versus the frequency shift. The orange solid line in the same figure represents Δk_W for (21,01→21,01), and the green solid line represents Δk_W for (21,11→21,11). Neglecting the nonlinear effect contribution Δk_{NL}, the intersections of the three solid lines and the red dashed curve are the phase-matching frequency shifts of the IMFWM for the three fiber mode combinations. Similarly, for Case 2 of the fiber mode combinations, we now consider (01,01→01,11), (01,01→01,21), and (11,11→11,21). According to Eq. (17), the blue solid line in Fig. 2(b) is the waveguide contribution Δk_W for (01,01→01,11) plotted versus the frequency shift. The orange solid line in the same figure represents Δk_W for (01,01→01,21), and the green solid line represents Δk_W for (11,11→11,21). The intersections of the three solid lines and the red dashed curve are the phase-matching frequency shifts of the three fiber mode combinations for Case 2.

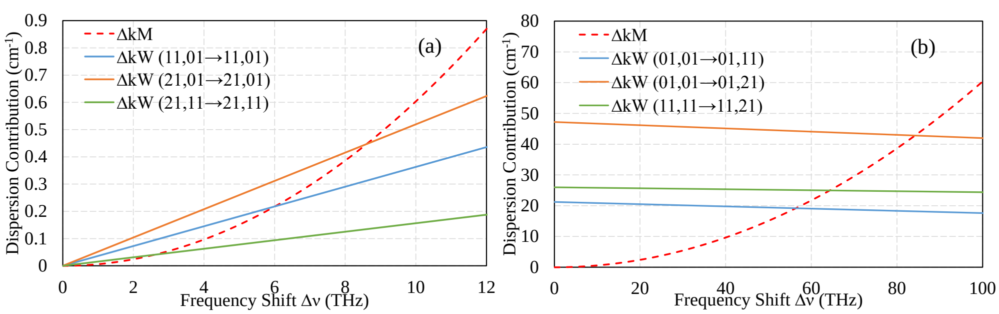

Fig. 2. Phase-matching diagrams of the IMFWM for the 25/400 fiber: (a) phase-matching diagram for Case 1 of the fiber mode combinations; and (b) phase-matching diagram for Case 2 of the fiber mode combinations.

The phase-matching frequency shifts of the IMFWM corresponding to different fiber mode combinations are listed in Table 1, and the Stokes-peak wavelength λ_3 and the anti-

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Stokes-peak wavelength λ_4 are calculated according to the phase-matching frequency shift. Then, assuming that the fluctuation in the core radius $\delta a = 0.01$ µm, the coherence lengths for different fiber mode combinations are calculated according to Eqs. (19) and (20) and listed in Table 1.

Table 1. Phase-matching Frequency Shift and Coherence Length for the 25/400 Fiber

Fiber mode combination	Case 1			Case 2		
	(11,01→ 11,01)	(21,01→ 21,11)	(21,11→ 21,11)	(01,01→ 01,11)	(01,01→ 01,21)	(11,11→ 11,21)
Phase-matching frequency shift /THz	6	8.6	2.6	56.4	84.2	64.3
Stokes-peak wavelength λ_3 /nm	1103.8	1114.5	1090.2	1355.1	1549.8	1405.3
Anti-Stokes-peak wavelength λ_4 /nm	1057.2	1047.6	1070	897.7	828.8	877
Coherence length /m	3335.5	40.8	137.6	2.59	0.43	1.77

As shown in Table 1, the coherence length for Case 1 is much larger than that of Case 2. Since the fiber length of a high-power fiber laser is generally longer than 10 m, which is much greater than the coherence length of Case 2, the IMFWM corresponding to Case 2 cannot be seen in high-power fiber lasers. Thus, in these lasers, it is only necessary to analyze the IMFWM corresponding to Case 1 of the fiber mode combinations.

3.2 Relationship between the fiber laser parameters and IMFWM

According to the calculation model, the phase-matching frequency shift is related to various fiber parameters, such as, the core radius, numerical aperture, and light wavelength, and fibers with different parameters support different numbers of fiber modes. An increase in the number of fiber modes can lead to an increase in the number of fiber mode combinations. As the phase-matching frequency shifts corresponding to different fiber mode combinations are different, the greater the number of fiber modes supported in the fiber, the more Stokes and anti-Stokes peaks can be generated by IMFWM. When the light wavelength is 1090 nm, the number of pairs of IMFWM peaks that can be produced in high-power fiber lasers corresponding to the core radius and numerical aperture fiber parameters is shown in Fig. 3(a). The results indicate that as the core radius and numerical aperture increase, the number of pairs of IMFWM peaks increases. In addition, the phase-matching frequency shift for (11,01→11,01) corresponding to the core radius and numerical aperture fiber parameters is shown in Fig. 3(b), and indicates that these parameters can also affect the phase-matching frequency shift of the IMFWM.

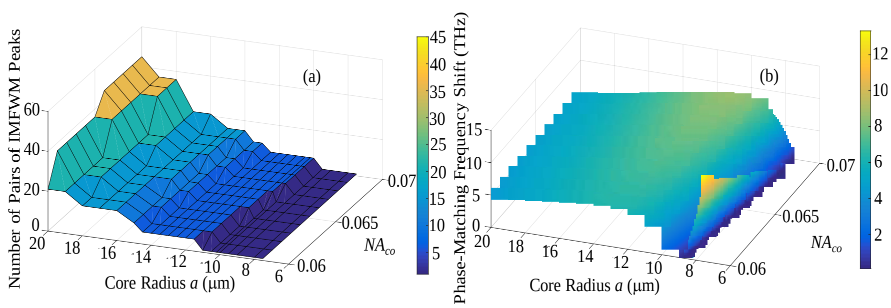

Fig. 3. Effect of the core radius and numerical aperture on the IMFWM: (a) the relationship between the core radius, numerical aperture, and the number of pairs of IMFWM peaks that can be produced in high-power fiber lasers; and (b) the relationship between the core radius, numerical aperture, and the phase-matching frequency shift of (11,01→11,01).

In addition, the light wavelength can also impact the IMFWM in high-power fiber lasers. An increase in the light wavelength can lead to a decrease in the normalized frequency, and thus resulting in a decrease in the number of pairs of IMFWM peaks that can be produced in high-power fiber lasers. Meanwhile, as the dispersion parameter β_2 and refractive index n of the fused silica are related to the light wavelength, variation in the light wavelength will result in a change in the phase-matching frequency shift. When the core radius a is 10 μm and the NA_{co} is 0.065, the number of pairs of IMFWM peaks that can be produced in high-power fiber lasers as the light wavelength increases from 1050 nm to 1100 nm are shown in Fig. 4(a). The results indicate that as the light wavelength increases, the number of pairs of IMFWM peaks decreases. For cases in which the number of pairs of IMFWM peaks is greater than 1, considering three fiber mode combinations (11,01→11,01), (21,01→21,01) and (21,11→21,11), the variations of the phase-matching frequency shifts with the light wavelength are shown in Fig. 4(b). For cases in which the number of pairs of IMFWM peaks is 1, the phase-matching frequency shift of (11,01→11,01) as the light wavelength increases from 1066 nm to 1100 nm are shown in Fig. 4(c). These results indicate that an increase in the light wavelength can lead to increases in the phase-matching frequency shifts of different fiber mode combinations.

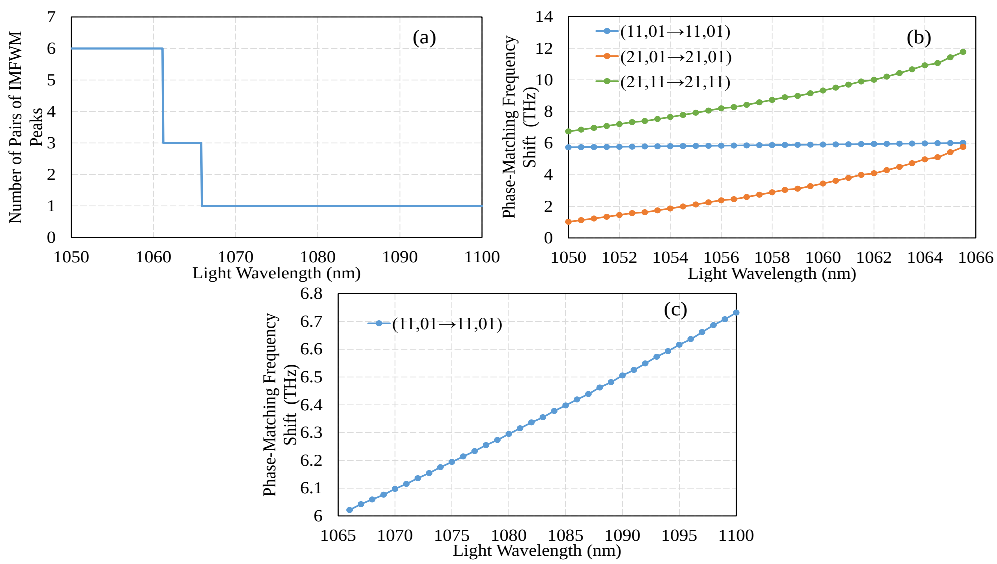

Fig. 4. Effect of the light wavelength on the IMFWM: (a) the relationship between the light wavelength and number of pairs of IMFWM peaks that can be produced in high-power fiber lasers; (b) the relationship between the light wavelength and the phase-matching frequency shift of three fiber mode combinations for cases in which the number of pairs of IMFWM peaks is greater than 1; and (c) the relationship between the light wavelength and the phase-matching frequency shift of (11,01→11,01) for cases in which the number of pairs of IMFWM peaks is 1.

3.3 Suppression of IMFWM in high-power fiber lasers

As shown in Figs. 3 and 4, the core radius, numerical aperture, and light wavelength parameters of the fiber have a great influence on the IMFWM in high-power fiber lasers. Variations in the laser parameters affect not only the phase-matching frequency shift, but also the number of pairs of IMFWM peaks. The IMFWM process results in deterioration of the output spectrum of the fiber laser and significant energy dispersion of the output laser, which limits its maximum obtainable output power and its application in the field of spectral beam combination. Therefore, it is desirable to suppress the IMFWM in high-power fiber lasers. According to the above theoretical analysis, optimizing the laser parameters is an effective way to suppress the IMFWM, and by selecting a fiber with small core radius and numerical aperture, and choosing a signal light of longer wavelength, the number of pairs of IMFWM

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peaks can be effectively reduced. However, decreases in the core radius and numerical aperture will limit improvements in the output power of the fiber laser, and the choice of the wavelength of the signal light must be made while considering its effect on the efficiency of the laser. Therefore, as suppressing the IMFWM by optimizing the laser parameters is somewhat limited, this method can only partially suppress the IMFWM in high-power fiber lasers.

The DCFs commonly used in high-power fiber lasers include the 20/400 [19], 25/400 [25], 30/400 [28], and 30/600 fibers [20]. Based on the results of analyzing the relationship between the fiber laser parameters and the IMFWM products, we selected 20/400 fiber with an NA_{co} of 0.06 for our testing, and a pair of FBGs with a center wavelength of 1090 nm as the laser cavity to suppress the IMFWM. However, the 20/400 fiber can still support two fiber modes (LP_{01} and LP_{11}) at the wavelength of 1090 nm, which indicates that the IMFWM can be generated in the high-power fiber laser used the 20/400 fiber.

In order to further suppress the IMFWM on the basis of laser parameter optimization, we propose a suppression method based on the mode control technique, which can be used to reduce the number of fiber modes propagating in the fiber, and thereby suppress the IMFWM. This technique includes the fiber coiling method, mode conversion method, fiber tapering method, and so on. However, compared with the other methods, the fiber coiling method is the simplest, most effective, and economical. When coiling the Yb-doped fiber in high-power fiber lasers, since the bend loss of high-order modes is greater than that of the fundamental mode, selecting the appropriate bending radius can reduce the power proportion of the high-order mode propagating in the fiber, and thereby suppress the generation of IMFWM products.

Taking a 20/400 Yb-doped fiber amplifier as an example, the influence of the bending radius on the output power of each mode was analyzed (via simulation) to demonstrate the possibility of suppressing the IMFWM by fiber coiling. In the simulation model, the bend loss of each mode in the fiber can be calculated using the bend loss formula. The simplified bend loss formula presented by Marcuse in Ref [29], when modified to include the fiber stress caused by the effective bend radius [30], can be written as

$$2\alpha = \frac{\sqrt{\pi}U^2 \exp(-2W^3 R_{eff} / 3a^3\beta^2)}{e_m \sqrt{aR_{eff}} W^{3/2} V^2 K_{m-1}(W) K_{m+1}(W)}. \tag{22}$$

Here, 2α is the power loss coefficient, and for silica fiber, the effective bend radius R_{eff} is about $1.28R$, where R is the bending radius of the fiber. The K terms are modified Bessel functions, and m is an azimuthal mode number corresponding to the subscript in LP_{mn}. For $m = 0$, $e_m = 2$, and for $m \neq 0$, $e_m = 1$.

Then, the bend loss of each mode as a function of the bending radius is substituted into a typical multimode fiber amplifier model [31] to analyze the relationship between the output power of each mode and the bending radius. The space-dependent and time-independent steady-state rate equations of multimode fiber amplifiers with fiber coiling can be expressed as

$$\frac{N_2(r,\varphi,z)}{N_1(r,\varphi,z)} = \frac{\dfrac{[P_p^+(z)+P_p^-(z)]\sigma_{ap}\Gamma_p(r,\varphi)}{h\nu_p} + \sum_i \dfrac{P_{si}^+(z)\sigma_{as}\Gamma_{si}(r,\varphi)}{h\nu_s}}{\dfrac{[P_p^+(z)+P_p^-(z)]\sigma_{ep}\Gamma_p(r,\varphi)}{h\nu_p} + \dfrac{1}{\tau} + \sum_i \dfrac{P_{si}^+(z)\sigma_{es}\Gamma_{si}(r,\varphi)}{h\nu_s}}, \tag{23}$$

$$\pm\frac{dP_p^\pm(z)}{dz} = \left\{\int_0^{2\pi}\int_0^a \left[\sigma_{ep}N_2(r,\varphi,z) - \sigma_{ap}N_1(r,\varphi,z)\right]\Gamma_p(r,\varphi) r dr d\varphi\right\} P_p^\pm(z) - \alpha_p P_p^\pm(z), \tag{24}$$

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$$\frac{dP_{si}^{+}(z)}{dz} = \left\{\int_0^{2\pi}\int_0^{a}\left[\sigma_{es}N_2(r,\varphi,z)-\sigma_{as}N_1(r,\varphi,z)\right]\Gamma_{si}(r,\varphi)rdrd\varphi\right\}P_{si}^{+}(z) - [\alpha_s+\alpha_{si}^{bend}(R)]P_{si}^{+}(z)-\sum_j d_{ij}[P_{si}^{+}(z)-P_{sj}^{+}(z)] , \quad (25)$$

where $N_1(r,\varphi,z)$ and $N_2(r,\varphi,z)$ are the population densities of the lower and upper lasing levels at the position (r,φ,z) respectively ($N(r,\varphi,z) = N_1(r,\varphi,z) + N_2(r,\varphi,z)$ is the doping concentration distribution); h is Planck constant; τ is the spontaneous lifetime of the upper lasing level; v_p and v_s are pump and signal frequencies; $P_p^{+}(z)$ and $P_p^{-}(z)$ are the pump powers in the forward and backward directions respectively; $P_{si}^{+}(z)$ is the signal power of the i^{th} transverse mode in the forward direction; σ_{ap} (σ_{ep}) and σ_{as} (σ_{es}) are the pump absorption (emission) and signal absorption (emission) cross-sections respectively; $\Gamma_p(r,\varphi)$ and $\Gamma_{si}(r,\varphi)$ are the power filling distributions of pump and signal of the i^{th} mode; α_p is the pump loss factor; α_s is the signal loss factor due to the fiber; $\alpha_{si}^{bend}(R)$ is the bend loss of the i^{th} mode as a function of the bending radius R calculated by Eq. (22); d_{ij} is the power coupling coefficient between the i^{th} mode and the j^{th} mode.

In order to simplify the calculation process and avoid complex two-dimensional integral operations, the multilayered method proposed by Gong et al in Ref [31]. is applied in our simulation to simplify Eqs. (23), (24), and (25). And then, the simplified rate equations of multimode fiber amplifiers can be solved under the boundary conditions which can be express as

$$P_p^{+}(0) = P_p^{f},\ P_p^{-}(L) = P_p^{b},\ P_{si}^{+}(0) = P_{si}^{in}. \quad (26)$$

Here, L is the fiber length; P_p^{f} is the input pump power in the forward direction; P_p^{b} is the input pump power in the backward direction; P_{si}^{in} is the initial signal power of the i^{th} mode. The parameters we used in the simulation of fiber coiling for a 20/400 Yb-doped fiber amplifier are shown in Table 2.

Table 2. Parameters Used for Simulations of a 20/400 Yb-doped Fiber Amplifier

Parameter	Value	Parameter	Value	Parameter	Value
λ_p	976 nm	N	7.94×10^{25} m^{-3}	P_p^{f}	1100 W
λ_s	1090 nm	τ	0.88 ms	P_p^{b}	0 W
σ_{ap}	1.7669×10^{-24} m^2	α_p	0.0035 /m	P_{s1}^{in} (LP_{01})	60 W
σ_{ep}	1.7131×10^{-24} m^2	α_s	0.0057 /m	P_{s2}^{in} (LP_{11})	20 W
σ_{as}	0.001×10^{-24} m^2	a	10 μm	L	10 m
σ_{es}	0.1895×10^{-24} m^2	NA_{co}	0.06	d_{ij}	0.005

The results of simulating the relationship between the output power of each mode and the bending radius are shown in Fig. 5, where it can be seen that as the bending radius of the gain fiber decreased, the output power of LP_{01} increased and the output power of LP_{11} decreased. These results indicate that the fiber coiling method can be used to effectively control the power proportion of high-order modes and reduce the number of fiber modes propagating in the fiber, thereby suppressing the IMFWM. However, it should be noted that if the bending radius continues to decrease, the bend loss of LP_{01} increases and the output power of LP_{01} gradually decreases. Therefore, it is preferable to avoid coiling the fiber using a smaller bending radius than necessary to reduce the loss of the fundamental mode.

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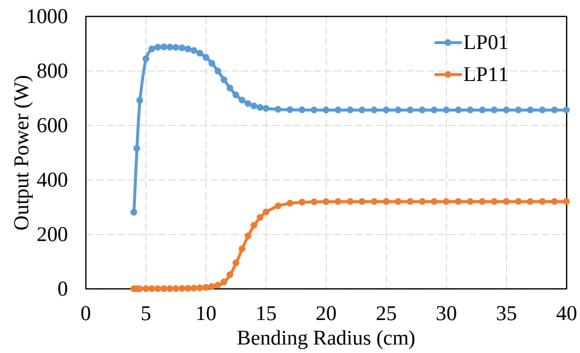

Fig. 5. Relationship between the output power of each mode and the bending radius for a 20/400 Yb-doped fiber amplifier.

4. Experiment

In order to verify the suppression of the IMFWM products by the fiber coiling method, a fiber laser was configured using the MOPA scheme shown in Fig. 6. The laser oscillator served as a seed laser, and included pump sources, a pump combiner (PC), a pair of FBGs, and gain fibers. The pump sources were laser diodes (LDs) near 976 nm that end-pump the laser oscillator. The high-reflector FBG (HRFBG) centered at a wavelength of ~1090 nm provided a 3-dB spectral bandwidth of about 1 nm with a reflective ratio more than 99%, and the output-coupler FBG (OCFBG) centered at a wavelength of ~1090 nm provided a 3-dB spectral bandwidth of about 0.06 nm with a reflective ratio about 11%. The gain fiber used in the laser oscillator was ~12 m of 20/400 μm double-clad ytterbium-doped fiber (YDF) with a numerical aperture of 0.06/0.46 (core/inner cladding). A cladding power stripper (CPS) was spliced after the laser cavity to remove the unabsorbed pump light and the signal light leaking into the inner cladding.

The output seed light of the laser oscillator was launched into the power amplifier through a pump-signal combiner (PSC). The gain fiber utilized in the power amplifier was ~12 m 20/400 μm double-clad YDF with a numerical aperture of 0.06/0.46. Another CPS was spliced after the power amplifier to remove unwanted light, and then a quartz block head (QBH) was utilized to deliver the output signal light of the power amplifier. The length of the passive fiber of the CPS and QBH was ~5 m. The output power of the power amplifier was measured by means of a power meter (Spiricon, 1500 W), and the output spectrum was measured using an optical spectrum analyzer (OSA, YOKOGAWA, AQ6370C).

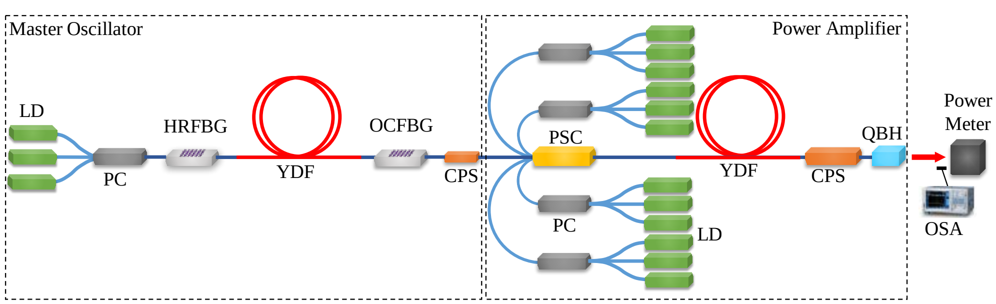

Fig. 6. Schematic of a fiber laser based on the MOPA scheme.

When the currents of laser diodes in the laser oscillator were set to 4 A, the pump power was 163 W, and the output power of the seed laser was 85.6 W. The output spectrum of the seed light is represented by the blue curve in Fig. 7(a). The output spectra at pump powers of 380, 760, and 1140 W are shown in Fig. 7(a) when the bending radius of the YDF in the fiber

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amplifier was 15 cm. The output power when different pump powers are launched into the fiber amplifier can be seen in Fig. 7(b).

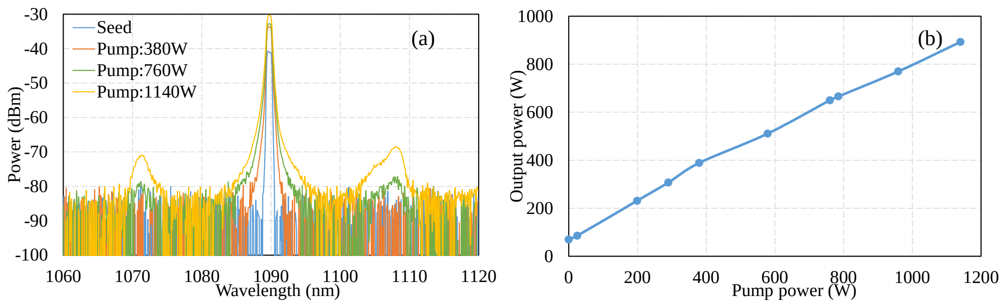

Fig. 7. Output characteristics of the power amplifier when the bending radius is 15 cm: (a) the output spectra at different pump powers; and (b) the output power versus the pump power.

As shown in Fig. 7(a), in addition to the main peak at 1090 nm, we can see two peaks at 1108 and 1071.6 nm, respectively, in the output spectra. According to the calculation model, the phase-matching frequency shift of (11,01→11,01) for the 20/400 YDF is 4.4 THz. Therefore, the theoretical values of the Stokes-peak wavelength λ_3 and anti-Stokes-peak wavelength λ_4 are 1107.7 and 1072.8 nm, respectively. Considering the calculation error due to the tolerance of the core radius and numerical aperture and the influence of the spectral width of the signal laser and the nonlinear effects, we believe that the two peaks observed in the output spectra are the Stokes and anti-Stokes peaks, respectively, produced by the IMFWM, and the corresponding fiber mode combination of this pair of IMFWM peaks is (11,01→11,01). Then, in order to suppress the IMFWM in this case, we employed the fiber coiling method to reduce the power proportion of the high-order modes.

According to the above theoretical analysis of the fiber coiling method, a decrease in the bending radius of the gain fiber can effectively reduce the power proportion of the high-order modes to suppress the generation of IMFWM. It is therefore essential to select the appropriate bending radius of the YDF. According to the simulation results shown in Fig. 5, we reduced the bending radius of the YDF to 7.5 cm. The output spectra of the fiber amplifier at the pump powers of 380, 760, and 1140 W are shown in Fig. 8(a). There were no peaks caused by IMFWM besides the main peak at 1090 nm in the output spectra.

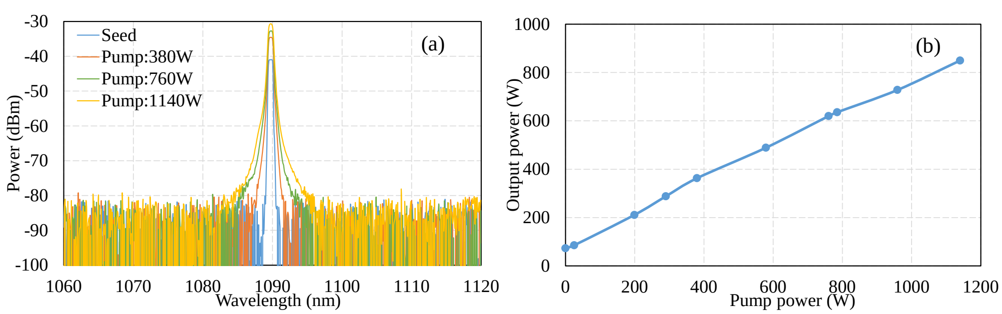

Fig. 8. Output characteristics of the power amplifier when the bending radius was 7.5 cm: (a) output spectra at different pump powers; (b) output power versus the pump power.

The experimental results indicate that the fiber coiling method can effectively suppress the generation of IMFWM products in high-power fiber lasers. In addition, comparing the output power shown in Figs. 7(b) and 8(b) indicates that reducing the bending radius can lead to an increase in the bend loss of LP_{11}, resulting in a decrease in the output power. When the pump power was 1140 W, the test result of output power was less than the simulation result shown

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in Fig. 5, which is due to the influence of factors such as splice losses, device losses, and spectral width of LDs.

5. Conclusions

In conclusion, we proposed a calculation model of the phase-matching frequency shift and coherence length to determine the fiber mode combination corresponding to each pair of IMFWM peaks observed in the output spectrum. We then analyzed the influence of the fiber core radius, numerical aperture, and light wavelength laser parameters on the generation of IMFWM products to provide theoretical guidance for optimizing the laser parameters to suppress the IMFWM products. Considering the limitations of the laser parameter optimization approach, the fiber coiling method was applied to further suppress the IMFWM products. And the simulation model of the fiber coiling method was established to demonstrate the efficacy of HOMs suppression and to provide theoretical guidance for the selection of the fiber bending radius. Combining the fiber coiling method with laser parameter optimization, we set up a fiber laser system based on a MOPA configuration and 20/400 fiber. Then, the IMFWM peaks at 1108 and 1071.6 nm in the output spectrum were removed by reducing the bending radius of the gain fiber from 15 to 7.5 cm based on the simulation model, which indicates that the fiber coiling method can effectively suppress the IMFWM products. The proposed calculation model can aid in the search for other ways to suppress the IMFWM products and can provide essential theoretical guidelines. In addition, due to the complexity of high-power fiber laser systems, laser parameters such as fiber length, pump power, seed power, and spectral width of seed light may also impact the IMFWM. In future studies, we plan to analyze the relationship between these parameters and IMFWM to further optimize the fiber laser parameters, and then further suppress the IMFWM products combining with the fiber coiling method.

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